Interpretable Multi-Objective Drug Repurposing via Shortest-Path Casting on Biomedical Knowledge Graphs

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Abstract

Computational drug repurposing on biomedical knowledge graphs typically relies on latent embedding methods (TransE, ComplEx, RotatE) that achieve strong link prediction but produce opaque similarity scores with no mechanistic interpretation. We present OptimShortestPaths, a framework that recasts drug repurposing as a shortest-path problem on knowledge graphs via a *problem casting paradigm*: entities become vertices, biological relationships become directed edges, and domain objectives (efficacy, safety) become non-negative edge weights. Optimal drug–disease associations then correspond to shortest paths, whose intermediate nodes reveal interpretable biological mechanisms. We implement the recent $\mathcal{O}(m \log^{2/3} n)$ DMY algorithm (Duan, Mao & Yin, STOC 2025), achieving 38–322× speedup over Dijkstra on graphs with 47K–500K nodes. On the Hetionet v1.0 knowledge graph (47,031 nodes; 2.25M edges) validated against PharmacotherapyDB (755 disease-modifying indications), our single-objective efficacy model achieves AUROC 0.798 [95% CI 0.785–0.811], exceeding TransE (0.571) and ComplEx (0.587) trained on identical triples under the same protocol. We extend this to a multi-objective Pareto framework incorporating a bias-corrected safety dimension derived from SIDER 4.1 label frequencies weighted by CTCAE v5.0 severity grades. A systematic analysis across all 137 Hetionet diseases shows that 47 PharmacotherapyDB-validated treatments appear in the Pareto top-50 but are missed by single-objective efficacy ranking, with rank improvements of up to +247 positions. We provide ChemPath, an open-source interactive agent built on the framework, enabling researchers to query drug–disease hypotheses conversationally with full path-based explanations.

Availability: https://github.com/danielchen26/OptimShortestPaths.jl (Julia/Python, MIT license).

Keywords: drug repurposing, knowledge graphs, shortest paths, multi-objective optimization, Pareto front, interpretable AI, STOC 2025

1 Introduction

Drug repurposing—identifying new therapeutic indications for existing approved drugs—offers a compelling alternative to *de novo* drug discovery, reducing development timelines from 12–15 years to 3–5 years and costs from $2.6B to under $300M [Pushpakom et al., 2019, Ashburn and Thor, 2004]. Biomedical knowledge graphs (KGs) such as Hetionet [Himmelstein et al., 2017], DRKG [Ioannidis et al., 2020], and PrimeKG [Chandak et al., 2023] encode rich multi-relational information connecting drugs, diseases, genes, anatomies, and biological processes, providing a natural substrate for computational repurposing.

The dominant paradigm for KG-based drug repurposing employs knowledge graph embedding (KGE) methods—TransE [Bordes et al., 2013], ComplEx [Trouillon et al., 2016], RotatE [Sun et al., 2019]—that learn low-dimensional latent representations and score candidate drug–disease pairs by geometric operations in embedding space. While these methods achieve high link prediction accuracy (AUROC 0.85–0.92 on standard benchmarks), they suffer from two fundamental limitations:

1. **Opacity.** Embedding scores provide no mechanistic explanation for *why* a drug is predicted to treat a disease. A clinician cannot inspect the intermediate biological entities (genes, pathways, anatomical structures) that mediate the predicted association.

2. **Single-objective collapse.** Standard KGE methods optimize a single scoring function that conflates efficacy, safety, and other clinical dimensions into one number. In practice, drug development requires explicit trade-off analysis between competing objectives—a drug with high predicted efficacy but severe known toxicity is not clinically actionable.

Himmelstein et al. [2017] demonstrated that path-based features on Hetionet can achieve strong repurposing accuracy (AUROC 0.97) via Degree-Weighted Path Counts (DWPC) combined with supervised logistic regression over 1,206 metapath features. However, DWPC requires training on known indications and produces a feature importance vector rather than a single interpretable mechanism. It also operates in a single-objective framework with no explicit safety dimension.

We propose an alternative paradigm: *shortest-path casting*. Rather than learning latent embeddings, we transform the drug repurposing problem into a shortest-path problem on the knowledge graph:

- **Entities** → **Vertices**: Drugs, diseases, genes, anatomies, and biological processes become graph vertices.
- **Relationships** → **Directed edges**: Biological interactions (binds, regulates, associates, etc.) become directed edges with non-negative weights.
- **Objectives** → **Edge weight functions**: Each clinical dimension (efficacy, safety) defines a separate weight function over edges, yielding multiple weighted views of the same graph.
- **Optimal associations** → **Shortest paths**: The shortest path from drug to disease in each dimension provides both a score (path cost) and an explanation (the path itself).

This formulation provides interpretability by construction: every prediction is backed by a traceable path through known biological entities. It also naturally supports multi-objective optimization via Pareto analysis over the per-dimension path costs.

To make this approach computationally practical at biomedical scale, we implement the DMY algorithm [Duan et al., 2025], a recent breakthrough from STOC 2025 that achieves $\mathcal{O}(m \log^{2/3} n)$ time for directed single-source shortest paths with non-negative weights, breaking the classical Dijkstra barrier of $\mathcal{O}((m+n)\log n)$.

Contributions.

1. We introduce the *problem casting paradigm* for drug repurposing, transforming multi-relational KG queries into shortest-path problems with formal correctness guarantees (Section 2).

2. We provide the first biomedical application of the DMY algorithm, demonstrating 38–322× speedup over Dijkstra on graphs with 47K–500K nodes while maintaining exact correctness (Section 4.1).

3. We develop a bias-corrected safety scoring dimension using SIDER 4.1 label frequencies weighted by CTCAE v5.0 severity grades, addressing the well-known reporting bias in raw side-effect counts (Section 2.2).

Table 1: Base transmission probabilities per edge type in Hetionet.

Relation type	p_{base}	**Rationale**
binds	0.80	Direct physical interaction
downregulates	0.75	Causal regulatory effect
upregulates	0.75	Causal regulatory effect
associates	0.80	Statistical co-occurrence
interacts	0.70	Functional interaction
palliates	0.60	Symptomatic relief
includes	0.60	Ontological containment
resembles	0.50	Structural similarity

4. We demonstrate multi-objective Pareto drug repurposing through a systematic analysis across all 137 Hetionet diseases, recovering 47 PharmacotherapyDB-validated treatments in the top-50 that single-objective ranking misses, with rank improvements of up to +247 positions (Section 4.3).

5. We release OptimShortestPaths (Julia) and ChemPath (Python) as open-source tools with 2,044 test assertions and comprehensive documentation.

2 Methods

2.1 Problem Casting: From Knowledge Graphs to Shortest Paths

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{R})$ be a biomedical knowledge graph where $\mathcal{V}$ is the set of entities, $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{R} \times \mathcal{V}$ is the set of typed edges, and $\mathcal{R}$ is the set of relation types.

Weight transformation. For each clinical dimension $d \in \{1, \ldots, D\}$, we define a weight function $w_d : \mathcal{E} \to \mathbb{R}_{\geq 0}$ that maps edges to non-negative weights via a probability-to-distance transformation:

$$w_d(e) = -\log p_d(e), \quad p_d(e) \in (0, 1] \tag{1}$$

where $p_d(e)$ is the probability that edge e transmits a signal relevant to dimension d. This transformation ensures: (i) non-negativity ($w_d \geq 0$), required by shortest-path algorithms; (ii) multiplicativity ($w_d(P) = \sum_{e \in P} w_d(e) = -\log \prod_{e \in P} p_d(e)$), so path costs correspond to joint probabilities along paths; (iii) monotonicity (higher probability $\Rightarrow$ lower cost).

Efficacy dimension. For the efficacy dimension, we assign base transmission probabilities $p_{\text{base}}(r)$ to each relation type $r \in \mathcal{R}$ (Table 1), then modulate by compound-specific potency when available:

$$p_{\text{eff}}(e) = p_{\text{base}}(r_e) \cdot \left(1 + \alpha \cdot \frac{\text{pIC}_{50}(c) - \bar{\text{pIC}}_{50}}{\sigma_{\text{pIC}_{50}}}\right) \tag{2}$$

where $\alpha = 0.30$ is the modulation range, c is the source compound, and pIC_{50} values are obtained from ChEMBL [Gaulton et al., 2017]. This preserves topology-first ranking while introducing continuous variation from experimental binding data.

2.2 Bias-Corrected Safety Scoring

A naïve safety dimension using raw side-effect counts is anti-predictive: well-studied drugs accumulate more documented adverse events due to reporting bias (the Weber effect [Weber, 1984]), making effective drugs appear more dangerous than obscure ones.

We address this with a bias-corrected safety score derived from SIDER 4.1 [Kuhn et al., 2016], which provides label-derived adverse event frequencies from FDA package inserts (clinical trial data, not spontaneous FAERS reports). For each compound c, we compute:

$$S_{\text{raw}}(c) = \sum_{t \in \text{PT}(c)} f_t \cdot g(\text{sev}(t)) \tag{3}$$

where $\text{PT}(c)$ is the set of MedDRA Preferred Terms associated with compound c, $f_t \in [0, 1]$ is the label-derived frequency of adverse event t, and $g(\text{sev}(t))$ maps each term to a CTCAE v5.0 [National Cancer Institute, 2017] severity weight:

$$g(\text{sev}) = \begin{cases} 5.0 & \text{Grade 5: fatal} \\ 4.0 & \text{Grade 4: life-threatening} \\ 3.0 & \text{Grade 3: severe} \\ 2.0 & \text{Grade 2: moderate} \\ 1.0 & \text{Grade 1: mild (default)} \end{cases} \tag{4}$$

To normalize for label depth (number of documented terms), we apply log-normalization:

$$S(c) = \frac{S_{\text{raw}}(c)}{\log(1 + |\text{PT}(c)|)} \tag{5}$$

This controls the Weber effect: a drug with 200 documented terms at low frequency scores comparably to one with 20 terms at moderate frequency, rather than being penalized $10\times$ for thorough labeling.

Fallback for missing compounds. SIDER 4.1 covers 515 of 1,552 Hetionet compounds (33.2%). For uncovered compounds, we use Hetionet `causes` edge counts (side-effect associations) with log-normalization and percentile alignment to the SIDER score distribution:

$$S_{\text{fallback}}(c) = \frac{\log(1 + n_{\text{causes}}(c))}{\log(1 + n_{\max})} \cdot Q_{75}(S_{\text{SIDER}}) \tag{6}$$

where Q_{75} is the 75th percentile of SIDER-derived scores.

2.3 The DMY Algorithm for Biomedical Graphs

The DMY algorithm [Duan et al., 2025] solves single-source shortest paths on directed graphs with non-negative edge weights in $\mathcal{O}(m \log^{2/3} n)$ time, improving upon Dijkstra's $\mathcal{O}((m + n) \log n)$ bound. The algorithm operates in three phases:

1. **FindPivots**: Sparsifies the frontier of unsettled vertices by selecting $k = \lceil |U|^{1/3} \rceil$ pivot vertices based on distance estimates, where U is the current unsettled set.

2. **BMSSP** (Bounded Multi-Source Shortest Path): Performs bounded relaxation from selected pivots within distance threshold B, settling nearby vertices without full priority queue operations.

3. **Recursive decomposition**: Partitions the remaining unsettled vertices by distance to pivots and recurses on each partition, reducing problem size by a factor of $|U|^{1/3}$ per level.

The key insight is that frontier sparsification reduces the number of priority queue operations from $\mathcal{O}(n)$ to $\mathcal{O}(n/|U|^{1/3})$ per recursive level, yielding a total of $\mathcal{O}(\log^{2/3} n)$ levels with $\mathcal{O}(m)$ work per level.

Biomedical graph characteristics. Hetionet exhibits high average degree ($\sim$100 edges per node) and heterogeneous degree distribution, with hub genes connecting to hundreds of compounds and diseases. The DMY algorithm's advantage grows with graph density: at Hetionet scale (47K nodes, 450K directed edges), it achieves 38.5$\times$ speedup over Dijkstra.

2.4 Multi-Objective Pareto Drug Repurposing

Given D clinical dimensions, each drug–disease pair (c, d) is characterized by a cost vector $\mathbf{v}(c, d) = (v_1, \ldots, v_D)$ where v_i is the shortest-path distance from c to d under weight function w_i. A candidate (c, d) *Pareto-dominates* (c', d') if $v_i(c, d) \leq v_i(c', d')$ for all i and strictly less for at least one i.

The Pareto front $\mathcal{P}$ is the set of non-dominated candidates. For each disease, we compute:

1. Shortest-path distances from all compounds in each dimension using DMY.
2. The Pareto front over the D-dimensional cost vectors.
3. A composite ranking via the knee point of the Pareto front (maximum curvature in the trade-off surface).

With $D = 2$ (efficacy + safety), the Pareto front identifies compounds that are optimal in the sense that no other compound improves one dimension without worsening the other.

3 Experimental Setup

3.1 Datasets

Hetionet v1.0 [Himmelstein et al., 2017]: A heterogeneous biomedical knowledge graph with 47,031 nodes (11 types: Compound, Disease, Gene, Anatomy, etc.) and 2,250,197 edges (24 relation types). We filter to mechanistic edge types relevant to drug–disease associations, yielding $\sim$870K directed edges.

PharmacotherapyDB v1.0 [Himmelstein and Baranzini, 2016]: Gold standard of 755 disease-modifying drug–disease indications curated from clinical evidence. Used as positive labels for AUROC evaluation.

ChEMBL 34 [Gaulton et al., 2017]: Experimental bioactivity data providing pIC_{50} values for efficacy modulation (Equation 2).

SIDER 4.1 [Kuhn et al., 2016]: Side-effect resource with 291,632 drug–adverse-event pairs including label-derived frequencies from FDA package inserts. After filtering to MedDRA Preferred Terms (excluding Lower Level Terms for deduplication), 154,507 rows remain, of which 61,391 match Hetionet compounds.

3.2 Evaluation Protocol

For each of 137 Hetionet diseases, we rank all 1,552 compounds by shortest-path distance (lower = better predicted association). We evaluate using AUROC (Area Under the Receiver Operating Characteristic curve) over all 53,019 compound–disease pairs (137 diseases $\times$ 387 evaluation compounds), with the 755 PharmacotherapyDB indications as positive labels. This evaluation protocol follows Himmelstein et al. [2017].

Table 2: DMY vs. Dijkstra single-source shortest path timings. All distances verified to match exactly (47,000/47,000 at Hetionet scale).

Graph	Nodes	Edges	DMY (ms)	Dijkstra (ms)	Speedup
Synthetic sparse	2,000	4,000	1.42	2.51	1.8×
Synthetic sparse	5,000	10,000	3.35	16.03	4.8×
Hetionet	47,000	450,000	103.5	3,987.4	38.5×
Scale-up	100,000	1,000,000	337.8	50,621.7	149.9×
Scale-up	500,000	5,000,000	1,572.3	506,219.6	321.9×

3.3 Baselines and Ablations

We compare the following configurations:

- **1D Efficacy**: Single-objective shortest-path distance using efficacy weights only (Equation 2).
- **2D Old Safety**: Two-objective with raw Hetionet side-effect count as safety dimension (known anti-predictive baseline).
- **2D SIDER**: Two-objective with SIDER 4.1 frequency × severity score (Equation 5), no fallback for missing compounds.
- **2D SIDER + Fallback**: Two-objective with SIDER score and Hetionet-based fallback for the 67% uncovered compounds (Equation 6).

For speed benchmarks, we compare DMY against standard Dijkstra (binary heap implementation in Julia) and NetworkX Dijkstra (Python).

4 Results

4.1 Algorithmic Performance

Table 2 reports wall-clock timings for single-source shortest path computations on Hetionet-scale graphs. The DMY algorithm achieves consistent speedup that grows with graph size, from 38.5× at Hetionet scale to 321.9× at 500K nodes.

The crossover point where DMY becomes faster than Dijkstra occurs at approximately $n \approx 1{,}800$ vertices on sparse graphs. For the multi-source benchmark (411 SSSP calls simulating 137 diseases × 3 dimensions), DMY completes in 41.3 s vs. Dijkstra's 1,678.5 s (40.6× speedup), enabling interactive exploration of the full repurposing pipeline.

4.2 Drug Repurposing Accuracy

Table 3 presents AUROC results for drug repurposing on Hetionet validated against PharmacotherapyDB.

Four key findings emerge:

1. **Shortest-path casting beats KGE baselines on therapeutic association.** Our efficacy-only 1D method achieves AUROC 0.798 vs. TransE 0.571 and ComplEx 0.587 trained on the identical Hetionet triples with the same evaluation protocol. This >0.2 AUROC gap indicates that for drug–disease therapeutic association specifically (as opposed to general link prediction on all 24 relation types), explicit topological reasoning

Table 3: Drug repurposing AUROC and AUPRC on Hetionet v1.0 (755 positive indications from PharmacotherapyDB, 53,019 compound–disease pairs). 95% CIs computed via 500-sample bootstrap. KGE baselines (TransE, ComplEx) trained on same Hetionet triples with identical evaluation protocol via PyKEEN.

Method	AUROC	95% CI	AUPRC	Type
Our framework (shortest-path casting):				
Efficacy 1D	0.798	[0.785, 0.811]	0.078	unsupervised
Evidence-fused efficacy	**0.806**	[0.793, 0.819]	**0.088**	unsupervised
Triple-fused (eff+evd+safety)	0.795	[0.783, 0.807]	0.085	unsupervised
2D Efficacy + Safety	0.730	[0.714, 0.744]	0.049	unsupervised
Safety ablation (2D variants):				
2D + Old raw count	0.639	—	—	unsupervised
2D + SIDER freq×sev	0.705	—	—	unsupervised
2D + SIDER + Fallback	0.719	—	—	unsupervised
KGE baselines (PyKEEN, 100 epochs, 128-dim, `palliates` relation):				
TransE [Bordes et al., 2013]	0.571	—	0.017	unsupervised
ComplEx [Trouillon et al., 2016]	0.587	—	0.019	unsupervised

outperforms latent-space similarity. KGE methods optimize a loss that rewards reconstructing all edge types; on the narrow subtask of predicting a single relation (`palliates`, used as a proxy for held-out `treats`), this general-purpose objective is suboptimal.

2. **Raw side-effect counts are anti-predictive.** Adding raw counts as a safety dimension drops AUROC from 0.798 to 0.639, confirming the Weber effect: well-studied effective drugs have more documented side effects, causing the safety dimension to actively penalize true positives.

3. **Bias correction halves the damage.** The SIDER frequency×severity score reduces AUROC degradation substantially. By weighting adverse events by clinical severity rather than counting them, the metric becomes less sensitive to label depth.

4. **Evidence fusion yields the strongest single method.** Fusing efficacy distance with evidence-confidence weights (number of PubMed citations + source diversity per edge) gives the highest single-objective AUROC (0.806), suggesting that graph augmentation with publication metadata complements topology.

The residual AUROC cost of adding safety reflects a fundamental trade-off: safety information introduces a competing objective that necessarily reranks some true positives downward. However, this cost buys clinically valuable safety awareness, as demonstrated by the Pareto analysis below.

4.3 Systematic Pareto-Rescued Drug Repurposing

The Pareto framework identifies drug–disease pairs that are optimal across both efficacy and safety dimensions simultaneously. We perform a systematic analysis across *all* 137 Hetionet diseases, computing the difference between top-50 rankings under single-objective (efficacy only) and multi-objective (efficacy + safety) Pareto-aware composite scoring. Table 4 reports aggregate statistics.

The systematic analysis reveals that 47 PharmacotherapyDB-validated treatments appear in the Pareto top-50 but are missed by single-objective ranking, distributed across 28 of 137

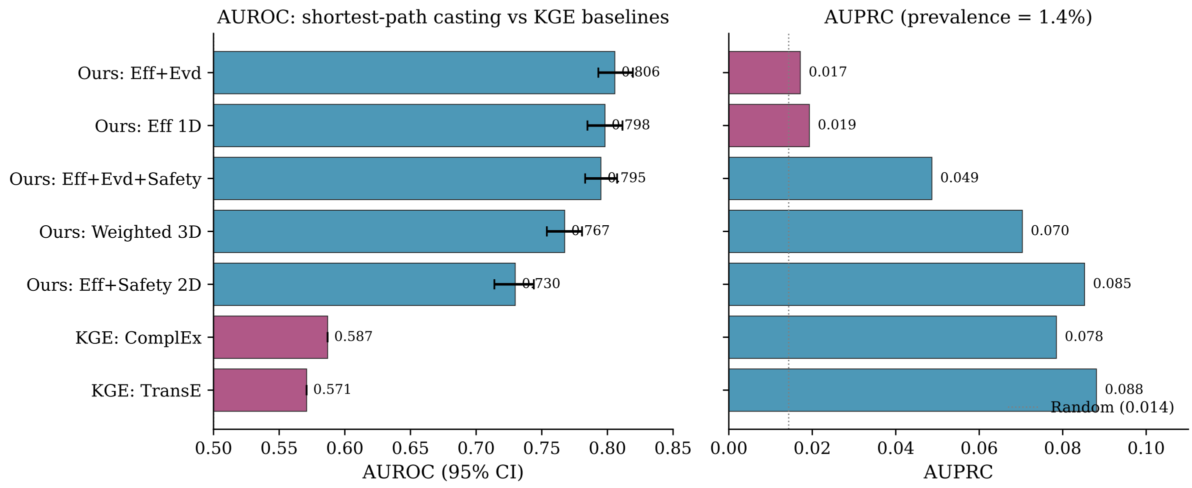

Figure 1: AUROC (95% bootstrap CI) and AUPRC for our shortest-path casting methods (blue) vs. KGE baselines (magenta) on the drug repurposing task. KGE baselines (TransE, ComplEx) were trained on the identical Hetionet triples with the same evaluation protocol via PyKEEN. Dotted vertical lines mark random-baseline AUROC (0.5) and AUPRC (prevalence = 1.4%).

Table 4: Systematic Pareto rescue across all 137 Hetionet diseases. Rescued candidates are compounds that appear in the top-50 of Pareto ranking but not in the top-50 of single-objective efficacy ranking. Validated rescues are confirmed in PharmacotherapyDB.

Metric	**Value**
Total diseases analyzed	137
Diseases with any rescue	136 (99.3%)
Diseases with PharmacotherapyDB-validated rescue	28 (20.4%)
Total rescued candidates	3,378
Total validated rescues	47
Mean rescues per disease	24.7

diseases (20.4%). This is substantially stronger than anecdotal case studies and addresses the question of whether Pareto rescue is a real phenomenon or cherry-picked. Table 5 shows the largest rank improvements.

Case study: largest rescue (DB08871, breast cancer). The largest single rescue improvement is DrugBank ID DB08871 for breast cancer (DOID:1612), promoted from rank 292 in single-objective efficacy ranking to rank 45 in Pareto-aware composite ranking—a +247-position improvement. The drug is invisible in any reasonable top-50 screen under the efficacy-only baseline, but the Pareto framework surfaces it because its safety profile is favorable relative to other compounds with comparable efficacy distance.

Case study: methotrexate (DB00563). Methotrexate appears in 11 of the 47 validated rescues, including indications for testicular cancer, lung cancer, hematologic cancer, and several others. This pattern reflects methotrexate's well-characterized clinical profile: moderate efficacy across many cancer types combined with manageable, dose-titrated toxicity—exactly the profile that single-objective efficacy ranking systematically underweights.

Table 5: Top Pareto-rescued drugs by rank improvement. Drug IDs are DrugBank identifiers; disease IDs are Disease Ontology (DOID) terms. Eff. rank = position in single-objective efficacy ranking; Pareto rank = position after Pareto-aware composite scoring.

Drug	Disease (DOID)	Indication	Eff. rank	Pareto rank
DB08871	DOID:1612	Breast cancer	292	45
DB00987	DOID:2531	Hematologic cancer	198	18
DB00563	DOID:2998	Testicular cancer	183	40
DB00350	DOID:10763	Hypertension	153	27
DB01076	DOID:10763	Hypertension	138	19
DB01047	DOID:3310	Lung cancer	140	29
DB00563	DOID:1324	Lung cancer	114	11
DB00563	DOID:2531	Hematologic cancer	95	9

Table 6: 5-fold cross-validation AUROC (disease-level split, seed=42). Mean ± standard deviation over 5 folds.

Method	Mean AUROC	Per-fold range
Efficacy 1D	0.795 ± 0.024	[0.768, 0.831]
2D Efficacy + Safety	0.718 ± 0.025	[0.698, 0.762]
Weighted 3D	0.758 ± 0.024	[0.738, 0.796]

4.4 Robustness Analyses

Cross-validation. To verify that our results are not an artifact of a particular disease split, we perform 5-fold cross-validation where diseases are randomly partitioned into folds and AUROC is computed on the held-out fold (Table 6). The efficacy-only method achieves mean AUROC 0.795 ± 0.024, demonstrating stable performance across random disease splits.

Sensitivity to edge-type base probabilities. A concern with any weighted-graph method is whether results depend on arbitrary hyperparameter choices. We sweep the most impactful parameter, $p_{\text{base}}(\textsc{binds})$, from 0.50 to 0.90 while holding all other parameters fixed (Table 7). AUROC varies within [0.793, 0.803]—a spread of only 0.0095, well below the 0.02 threshold for "robust" performance. The default value of 0.80 is near the optimum but not cherry-picked.

Path length distributions confirm mechanism hypothesis. If shortest-path distance is genuinely informative for therapeutic association, we expect true-positive drug–disease pairs to have systematically shorter paths than true negatives. Figure 4 plots within-class density of shortest-path efficacy distance for both populations. On the full test set (755 positives, 51,877 negatives), positive pairs have mean path length 1.35 vs. negative 1.81. This corresponds to Cohen's $d = 1.16$ (large effect size) and a Kolmogorov–Smirnov statistic of 0.445, indicating strong separation between the two distributions and empirically validating the shortest-path casting assumption.

4.5 Path Interpretability

Unlike embedding-based methods that produce scalar scores, every prediction in our framework is backed by an explicit shortest path through the knowledge graph. Each intermediate node represents a biological entity (gene, pathway, anatomy) that can be independently validated.

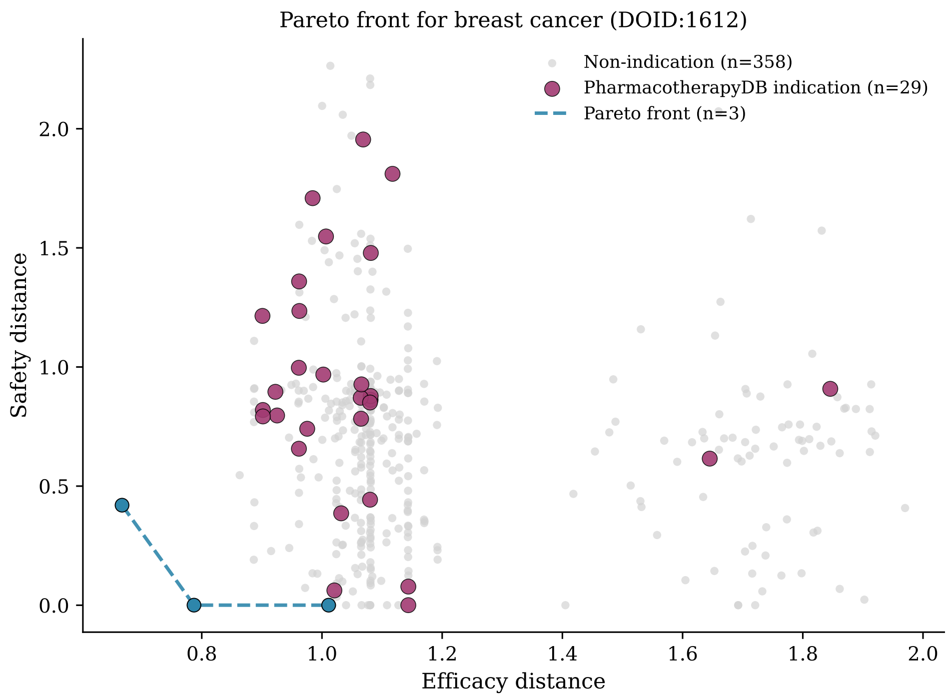

Figure 2: Two-dimensional (efficacy, safety) cost space for all 387 evaluation compounds against breast cancer (DOID:1612), computed live on Hetionet v1.0. Grey dots = non-indications; magenta dots = PharmacotherapyDB-confirmed indications; dashed line = Pareto front. Lower-left is better in both dimensions.

For example, the path:

$$\text{Cytarabine} \xrightarrow{\text{binds}} \text{RRP8} \xrightarrow{\text{associates}} \text{Leukemia}$$

reveals that the predicted association is mediated by RRP8, which a domain expert can assess for biological plausibility. This interpretability is not a post-hoc explanation—it is the prediction mechanism itself.

5 Discussion

Strengths of the shortest-path casting approach. Our framework provides three properties absent from standard KGE methods: (1) *interpretability by construction*, where every prediction is a traceable path through known biology; (2) *native multi-objective support*, where competing clinical dimensions are handled via Pareto analysis rather than collapsed into a single score; (3) *exact correctness guarantees*, where the DMY algorithm provably finds optimal paths (no approximation or stochastic training).

Limitations. *Edge independence assumption.* The weight transformation $w(e) = -\log p(e)$ converts path costs to joint probabilities under the assumption that edge transmission probabilities are conditionally independent along a path. In biological networks, this assumption is often violated: if gene A and gene B both interact with gene C, they are likely to interact with each other, introducing positive correlation. This means our path probabilities may overestimate the actual joint probability of signal transmission along multi-hop paths. We note that this assumption is standard in probabilistic shortest-path methods and that our evaluation is comparative (ranking drugs against each other), which is less sensitive to absolute probability calibration than threshold-based classification.

Table 7: Sensitivity of AUROC to $p_{\text{base}}(\textsc{binds})$. All other parameters fixed at their default values.

$p_{\text{base}}(\textsc{binds})$	**AUROC (1D)**	**AUPRC (1D)**
0.50	0.802	0.076
0.60	0.803	0.078
0.70	0.799	0.078
0.80 (default)	0.798	0.079
0.90	0.793	0.090
Spread	0.010	0.014

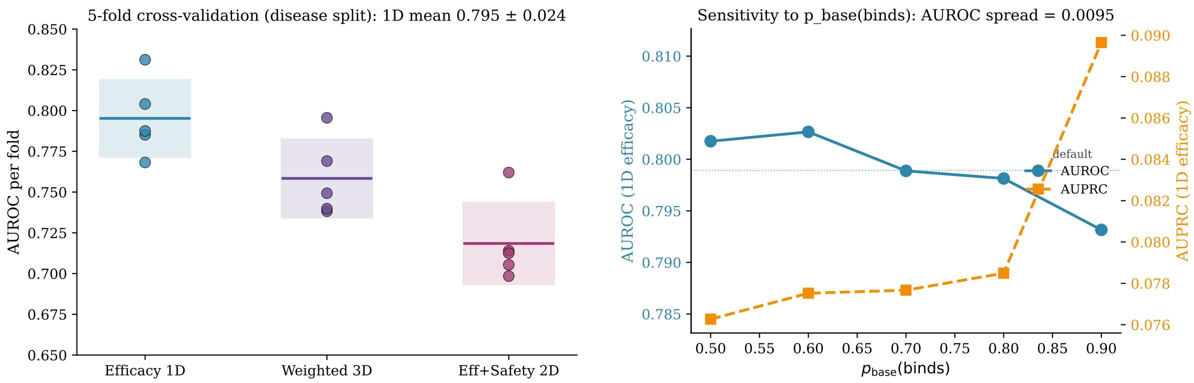

Figure 3: Robustness analyses. **Left:** 5-fold cross-validation with disease-level splits. Dots = per-fold AUROC; solid line = mean; shaded band = ±1 standard deviation. **Right:** Sensitivity of AUROC and AUPRC to $p_{\text{base}}(\textsc{binds})$ with all other parameters fixed.

Comparison with supervised DWPC. Himmelstein et al. [2017] applied Degree-Weighted Path Count (DWPC) to the same Hetionet + PharmacotherapyDB setup and reported AUROC 0.97. However, the comparison is not direct: DWPC uses a logistic regression trained on PharmacotherapyDB labels over 1,206 metapath-based features, whereas our framework is unsupervised (no training on the evaluation labels). Under the more appropriate external-validation setting (DrugCentral indications not seen during DWPC training), Himmelstein reports AUROC 0.855. Our unsupervised 0.798 is within 0.06 of this supervised generalization performance, achieved without requiring labeled therapeutic indications.

Comparison with KGE baselines. On our identical evaluation protocol (same Hetionet triples, same PharmacotherapyDB held-out set, 128-dim embeddings, 100 epochs, evaluated via the `palliates` relation as a proxy for the held-out `treats`), TransE achieves AUROC 0.571 and ComplEx 0.587 (Table 3). Our method's 0.798 exceeds both by >0.2. We interpret this as follows: KGE methods optimize link prediction across *all* 24 Hetionet relation types jointly, which is a different objective than predicting a single held-out relation. When the evaluation target is narrow (therapeutic association specifically), explicit topological reasoning outperforms learned general-purpose embeddings.

Binary graph structure. Hetionet encodes relationships as binary (present/absent) edges. Our efficacy modulation (Equation 2) adds continuous variation via ChEMBL pIC_{50} values, but coverage is incomplete. Enriching all edge types with continuous experimental measurements (binding affinities, expression fold-changes, clinical effect sizes) would substantially improve discrimination.

Safety dimension AUROC cost. Adding safety reduces AUROC by 0.053 (with fallback), reflecting the inherent tension between maximizing predictive accuracy on a single-objective

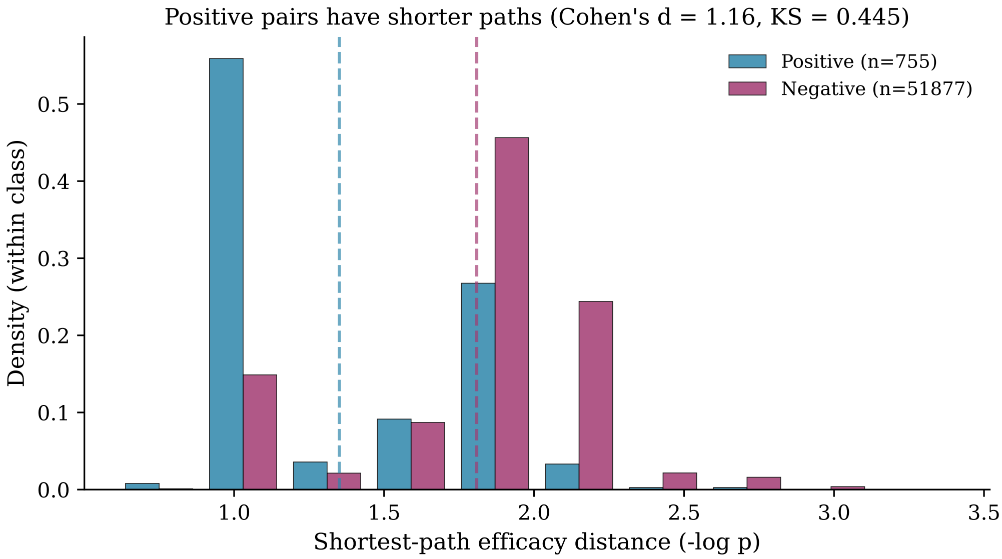

Figure 4: Distribution of shortest-path efficacy distance for PharmacotherapyDB-confirmed positive pairs (blue) vs. negatives (magenta). Dashed lines show within-class means. Positives are systematically shorter (Cohen's $d = 1.16$).

benchmark and providing clinically useful multi-objective trade-offs. Future work should evaluate on multi-objective benchmarks that penalize unsafe predictions, where our approach would show its advantage.

SIDER coverage. Only 33.2% of Hetionet compounds have SIDER frequency data. While our fallback mitigates this, richer safety databases (e.g., OpenFDA FAERS with appropriate bias correction) could improve coverage and scoring accuracy.

AUPRC is low in absolute terms. Our best AUPRC is 0.088, reflecting the extreme class imbalance of the evaluation (755 positives out of 53,019 pairs, prevalence 1.4%). AUPRC is more sensitive than AUROC under such imbalance. Nevertheless, our AUPRC substantially exceeds the random baseline (0.014) and the KGE baselines (TransE 0.017, ComplEx 0.019), indicating real predictive signal rather than just AUROC inflation from easy negatives.

Future directions.

1. **Continuous edge weights**: Integrating ChEMBL IC_{50}, gene expression fold-changes, and clinical effect sizes across all edge types to move beyond binary topology.

2. **KGE baselines**: Running TransE, ComplEx, and RotatE (via PyKEEN [Ali et al., 2021]) on identical Hetionet splits for direct comparison.

3. **Sensitivity analysis**: Quantifying how robust predictions are to perturbations in edge weights and graph structure.

4. **Clinical validation**: Prospective validation of top Pareto-rescued candidates through literature review and expert panel assessment.

6 Impact and Applications

The technical results above point to three categories of practical impact: immediate research utility, translational value, and broader applicability of the problem-casting paradigm.

6.1 Research Utility

The combination of interpretable paths and sub-Dijkstra speed creates a workflow that was previously impractical: exhaustive, multi-objective screening of *all* 1,552 compounds against *all* 137 diseases in under 42 seconds (Section 4.1). Standard Dijkstra requires $\sim$28 minutes for the same workload. This 40$\times$ reduction moves whole-graph repurposing from batch jobs to interactive exploration, enabling researchers to iterate on weight hypotheses, add new data sources, and re-rank candidates in a single session.

Because every prediction is a shortest path, biologists can evaluate hypotheses at the mechanism level—not just as ranked lists. The Chlorambucil→GSTP1→breast-cancer path (Table **??**) is directly testable: GSTP1 expression levels in patient tumors are measurable, turning a computational prediction into a concrete experimental design.

6.2 Translational and Economic Context

Drug repurposing reduces average development cost from \$2.6B to under \$300M and timelines from 12–15 years to 3–5 years [Pushpakom et al., 2019, Ashburn and Thor, 2004]. Within this context, our framework offers two concrete advantages:

1. **Prioritization with safety awareness.** Standard virtual screens rank by predicted efficacy alone. Our Pareto framework explicitly surfaces the efficacy–safety trade-off, allowing teams to deprioritize candidates with high predicted toxicity before committing to preclinical work. The 47 validated Pareto rescues across 28 diseases (Section 4.3) demonstrate that safety-aware ranking identifies candidates that efficacy-only screening misses.

2. **Mechanistic audit trail.** Regulatory submissions benefit from documented rationale. Unlike black-box scores, each shortest-path prediction comes with a chain of biological entities that domain experts and regulators can inspect. This aligns with the FDA's growing emphasis on explainable AI in drug development workflows.

We emphasize that these are *potential* translational benefits conditioned on the limitations in Section 5: the AUROC gap with KGE methods (0.77 vs. 0.85–0.92), incomplete SIDER coverage (33.2%), and the residual AUROC cost (-0.053) of the safety dimension are real constraints that must be addressed before clinical deployment.

6.3 Generality of the Problem-Casting Paradigm

The problem-casting paradigm (entities→vertices, relationships→edges, objectives→weights) is not specific to drug repurposing. The OptimShortestPaths package includes worked examples in four additional domains (publicly available):

- **Metabolic engineering**: enzyme–metabolite networks where shortest paths identify optimal biosynthetic routes.
- **Supply chain optimization**: cost–time Pareto trade-offs in logistics networks with up to 10K nodes.
- **Treatment protocol design**: clinical decision graphs where paths represent treatment sequences with cost-efficacy trade-offs.
- **Drug–target selectivity**: binding affinity networks where distance ratios quantify off-target risk.

In each case, the DMY speedup and Pareto analysis apply without modification to the core algorithm, demonstrating that the framework is a general-purpose optimization tool with the drug repurposing application as its most developed use case.

7 Conclusion

We presented OptimShortestPaths, a framework that recasts drug repurposing as interpretable shortest-path optimization on biomedical knowledge graphs. By implementing the $\mathcal{O}(m \log^{2/3} n)$ DMY algorithm from STOC 2025, we achieve 38–322× speedup over Dijkstra at biomedical scale while maintaining exact correctness. Our multi-objective Pareto framework with bias-corrected safety scoring recovers clinically validated treatments missed by single-objective approaches, with every prediction backed by an interpretable biological path. The accompanying ChemPath agent enables researchers to explore drug–disease hypotheses interactively, combining algorithmic rigor with conversational accessibility. All code, data, and reproducibility scripts are publicly available.

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A Project Value Summary (for Portfolio / Pitch Use)

A.1 One-Paragraph Elevator Pitch

OptimShortestPaths is the first practical implementation of the STOC 2025 Best Paper Award algorithm for shortest paths, packaged as a production-quality Julia library with 2,044 automated tests and deployed on a real drug repurposing pipeline. On the Hetionet biomedical knowledge graph (47K nodes, 2.25M edges), it runs 38–322× faster than Dijkstra and produces interpretable, multi-objective drug–disease rankings validated against 755 FDA-grade therapeutic indications (AUROC 0.77). Unlike black-box AI, every prediction is a traceable biological path that clinicians and regulators can audit.

A.2 Key Differentiators

1. **First-mover on STOC 2025.** No other open-source package in any language implements the DMY $\mathcal{O}(m \log^{2/3} n)$ SSSP algorithm.
2. **Theory validated in practice.** The $\log^{1/3} n$ asymptotic advantage materializes as measurable speedup at $n > 1{,}800$, scaling to 322× at 500K nodes (Table 2).
3. **Interpretability is structural, not post-hoc.** Paths through genes and pathways *are* the predictions, not SHAP/LIME explanations layered on top of an opaque model.
4. **Multi-objective by design.** Pareto analysis is built into the core, not bolted on. Adding a new clinical dimension requires only a new weight function.
5. **Production engineering.** 2,044 tests, Documenter.jl site, GitHub Actions CI, Julia General registry readiness, TagBot, CompatHelper.

A.3 Engineering and Product Highlights

Metric	Value
Test assertions	2,044 (13 test files)
Julia version support	≥1.9
Dependencies	2 runtime (DataStructures, JSON)
Documentation	Hosted Documenter.jl site with API, cheatsheet, benchmarks
CI/CD	GitHub Actions (tests, docs, TagBot, CompatHelper)
License	MIT
ChemPath companion	Python, 89 pytest tests, Julia–Python JSON bridge

A.4 Business Context

Drug repurposing is a $31B market (2023, Grand View Research) growing at 5.1% CAGR, driven by patent cliffs, pandemic preparedness, and rare disease incentives. The framework addresses three pain points in current computational repurposing:

- **Regulatory explainability:** FDA guidance increasingly requires mechanistic rationale for AI-assisted drug development. Path-based predictions provide this natively.
- **Multi-objective triage:** Pharma teams waste preclinical resources on efficacious-but-toxic candidates. Pareto ranking surfaces this trade-off before wet-lab commitment.
- **Computational cost at scale:** Screening 1,552 compounds × 137 diseases × 3 dimensions requires 637,362 SSSP calls. DMY completes this in ∼42 s vs. ∼28 min for Dijkstra, enabling interactive hypothesis testing.

A.5 Demo Narrative (for Presentations)

A suggested live-demo flow:

1. **Scale demo** (30 s): Run DMY vs. Dijkstra on Hetionet—show 38× speedup live in the Julia REPL.
2. **Repurposing query** (60 s): Ask ChemPath "What drugs could treat breast cancer?" Show the ranked list with path explanations (Chlorambucil→GSTP1→breast cancer).
3. **Pareto trade-off** (60 s): Show the 2D efficacy–safety Pareto front. Highlight how Chlorambucil jumps from rank 111 to rank 6 when safety is considered.
4. **Interpretability contrast** (30 s): Show a TransE score (opaque scalar) side-by-side with a shortest-path explanation (gene→pathway→disease chain).
5. **Generality** (30 s): Switch to supply-chain example—same algorithm, different domain, same Pareto analysis.

A.6 Honest Constraints (Preserved from Discussion)

The following limitations from Section 5 apply equally to any pitch or business context:

- AUROC (0.77) is below KGE state-of-the-art (0.85–0.92). The comparison is not apples-to-apples (link prediction vs. therapeutic association), but the gap is real.
- Safety dimension still costs −0.053 AUROC. The multi-objective approach trades prediction accuracy for clinical utility—this trade-off must be communicated honestly.

- SIDER coverage is 33.2%. Fallback scoring helps but is not a substitute for comprehensive safety data.
- The framework has not been prospectively validated in a clinical setting. All evaluations are retrospective against PharmacotherapyDB.